Name: _____ Period: _____

Unit 2 Park Tour: Matter and Energy in Our National Parks

Learning Intention: Today we will be learning how the ideas from Unit 2 show up in the national parks we studied so we can explain how energy and matter shape the places worth protecting.

Success Criteria:

- I can identify evidence of heat transfer, phase changes, and rock formation in a video clip.
- I can explain how each Day 1 rock formed using its rock type.
- I can use evidence to explain why a park is worth protecting.

Word Bank: conduction • convection • density • phase change • magma • lava • crystals • igneous • sedimentary • metamorphic • weathering • erosion • deposition • uplift • rock cycle

Key Terms:

- Weathering: Rock breaks into smaller pieces.
- Erosion: Water, wind, ice, or gravity carries broken rock away. Over time, a river can erode deep down into rock.
- Deposition: Moving water, wind, or ice drops the sediment it was carrying, and layers build up over time.
- Uplift: Forces inside Earth push large areas of land higher.

Part A: Energy in the Parks

Yellowstone is the only park today that is about energy instead of rocks.

Park and Question	What I Observe	My Answer
1. Yellowstone Where does the energy that heats the water come from? What phase change causes a geyser to erupt?		

Part B: Rocks from Day 1

On Day 1, you observed rocks from four national parks without naming them. Now you will name each rock type and explain how it formed. Use the video clips, the rock samples, and your notes.

Park and Day 1 Rock	What I Observe	My Answer
1. Hawai‘i Volcanoes What phase change happens as lava cools? What type of rock does it form?		
2. Yosemite Granite is igneous, like Hawai‘i's lava rock, but it formed in a different place. Where did it cool, and how do its crystals show that?		
3. Badlands What type of rock makes up the Badlands layers? What process is wearing them away?		
Great Smoky Mountains Compare this rock (slate and phyllite) to the Badlands shale. How is it different?		

Part C: Ocean Fossils in the Desert

Scientists have found ocean fossils in the walls of the Grand Canyon, a mile above sea level, in the middle of the desert. Use evidence from the video to explain how they got there.

Question	Evidence From the Video
1. How did the rock layers and fossils form? *(deposition)*	
2. How did the layers end up a mile above sea level? *(uplift)*	
3. How can we see the layers today? *(weathering and erosion)*	
Put it together: In one or two sentences, explain how ocean fossils ended up a mile above sea level.	

Part D: What Makes a Place Worth Protecting? (Revisited)

On Day 1, you argued that a park was worth protecting using only what you could observe. You did not know what the rocks were or how they formed. Now you do. Make the argument again, using the science from this unit.

#	Criterion	Ask yourself
1	**Outstanding example**	**Is this one of the best examples of this kind of place or feature anywhere in the country?**
3	**Outstanding for enjoyment or scientific study**	**Can people learn something here that they could not learn as well anywhere else?**

Brainstorm: Why Is Each Park Worth Protecting?

Jot quick notes for each park. Words or phrases are fine.

Park	Rock type	How it formed	What makes it special?
Yosemite			
Hawai‘i Volcanoes			
Badlands			
Great Smoky Mountains			
Which park has the strongest case? Circle it above.			

CER Formula:

C = Claim: Answer the question in one sentence.
E = Evidence: The data or observations that support your claim. *In the data/video/lab, I saw* _____.
R = Reasoning:

1. **The science rule:** *This happens because* _____. (A science idea that explains the evidence.)
2. **Connect back:** *This shows that* _____. (Tie it back to your claim.)

Check yourself: Reasoning explains *why* or *how* and uses a science word. If it only repeats the evidence, it isn't reasoning yet.

See the CER formula in action: The Day 1 answer only describes what you can see. Today's answer explains the science and connects it to Criterion 1 and 3. Your answer should look like the one on the right.

EXAMPLE: What does "using science" look like?	
Day 1 answer (observation only)	**Today's answer (using science)**
White Sands is worth protecting because the sand is bright white and the dunes are huge.	White Sands is worth protecting because it is the largest gypsum dunefield in the world (Criterion 1). Water carried dissolved gypsum into a lake. When the water evaporated, a phase change from liquid to gas, gypsum crystals were left behind. Wind broke the crystals into sand and deposited them as dunes. Scientists can study phase change, crystal formation, and deposition all in one place (Criterion 3).

This example shows the pattern: name the science, explain how it happened, then connects it to the criteria.

Directions: Choose one of the four Day 1 parks. Argue that it passes Criterion 1 and Criterion 3. Your answer must include:

- A rock type, process, or energy idea from Unit 2 (for example: igneous, convection, phase change, deposition, uplift, erosion)
- How that process happened at your park

Question: *Why is your park worth protecting? Does it pass Criterion 1 and Criterion 3?*

Claim: _____ is worth protecting because _____

_____

Evidence (something you saw in the video, rock sample, learned throughout the unit):

_____

_____

Reasoning (connect your evidence to a Unit 2 idea. Ask yourself: What science explains what I saw? Why is that hard to see or study somewhere else?):

_____

_____

_____

_____

_____

CER RUBRIC		
	1 point	**0 points**
Claim	Names one park and says why it's worth protecting	Missing, or names no park
Evidence	Something specific from the video, rock sample, etc.	Missing, or general ("it's pretty")
Reasoning: Science	Explains how or why it happened, using a Unit 2 idea correctly	Missing, wrong, or repeats the evidence
Reasoning: Criterion	Connects to Criterion 1 and 3	Missing one or both criteria